

CHAPTER – 6

MUCOCUTANEOUS DISEASES

LEARNING OUTCOMES

By the end of this chapter students will be able to:

- 1. Define the Muco-Cutaneous Lesions.**
- 2. Describe the Pathophysiology of Autoimmune Diseases.**
- 3. Explain the clinicopatholoical features of different mucocutaneous lesions.**

MUCOCUTANEOUS LESIONS

Muco-Cutaneous Lesions

Surface lesions in Mucous Membranes and Skin

Vesiculobullous
Diseases

Ulcerative
Conditions



White Lesions

Red-Blue
Lesions

Pigmented
Lesions

Verrucal-
Papillary
Lesions



1. Vesiculobullous Diseases

- Herpes Simplex Virus
- Herpes Zoster
- Herpangina
- Measles
- Pemphigus
- Pemphigoid
- Dermatitis Herpetiform
- Epidermolysis Bullosa



2. Ulcerative Conditions

- Reactive Lesions
- Contact Allergy
- Tuberculosis
- Aphthous Ulcers
- Erythema Multiforme
- Lupus Erythematosus
- Granulomatous Diseases
- Oral Squamous Cell Carcinoma



3. White Lesions

- Leukoedema
- Focal Hyperkeratosis
- Leukoplakia
- Geographic Tongue
- Lichen Planus
- Candidiasis
- Mucosal Burns
- Submucous Fibrosis

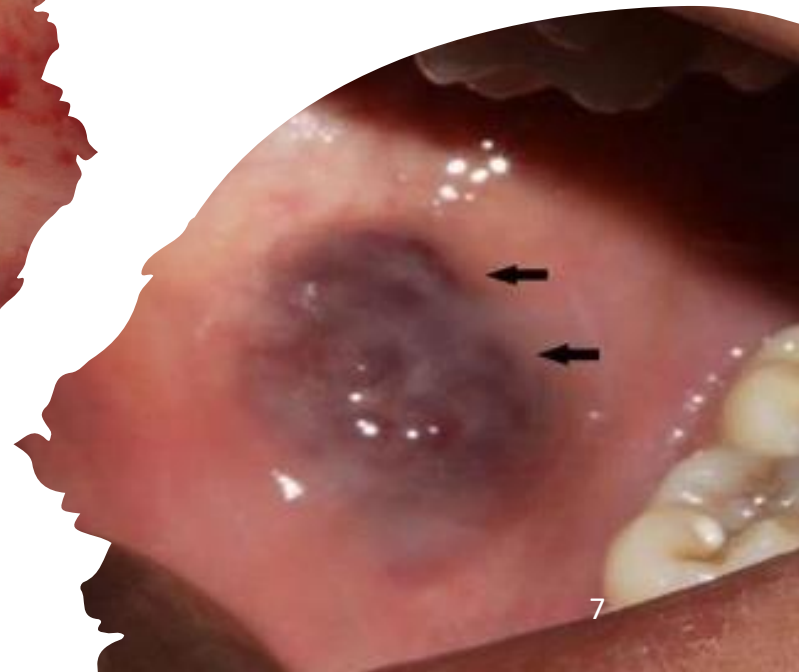
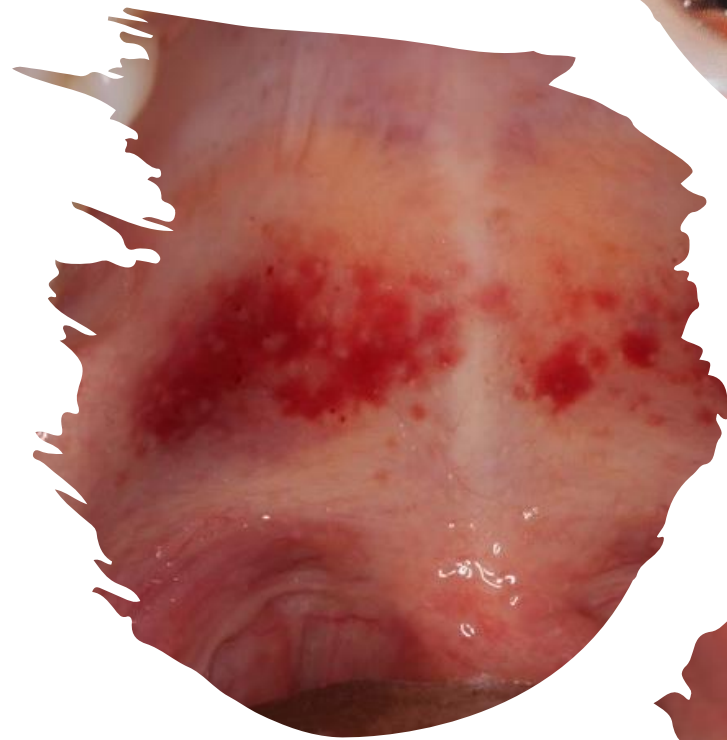


thick or verrucous



4. Red-Blue Lesions

- Blood Dyscrasias
- Hemangiomas & Vascular Malformations
- Pyogenic Granuloma
- Erythroplakia
- Vitamin-B Deficiency
- Anemia
- Scarlet Fever
- Traumatic Lesions



5. Pigmented Lesions

- Physiologic Pigmentation
- Smokers Melanosis
- Melanotic Macule
- Melanocytic Nevus
- Heavy Metal Pigmentation
- Drug Induced Pigmentation
- Melanoma
- Amalgam Tattoo



6. Verrucal-Papillary Lesions

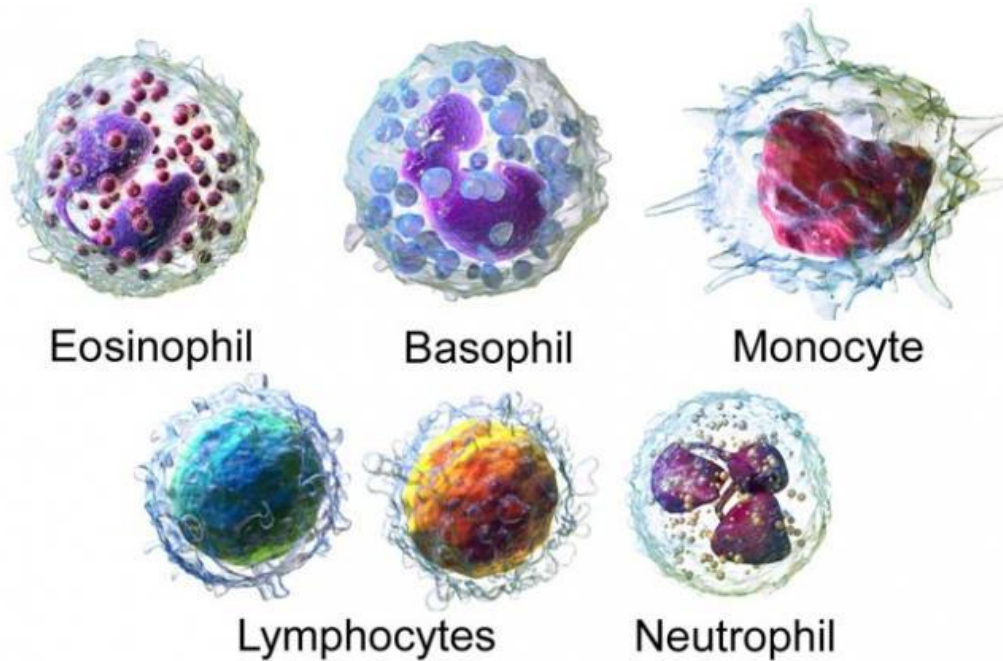
- Papillary hyperplasia
- Squamous papilloma
- Verruca vulgaris
- Condyloma acuminatum
- Focal epithelial hyperplasia
- Keratoacanthoma
- Verrucous carcinoma
- Verruciform xanthoma



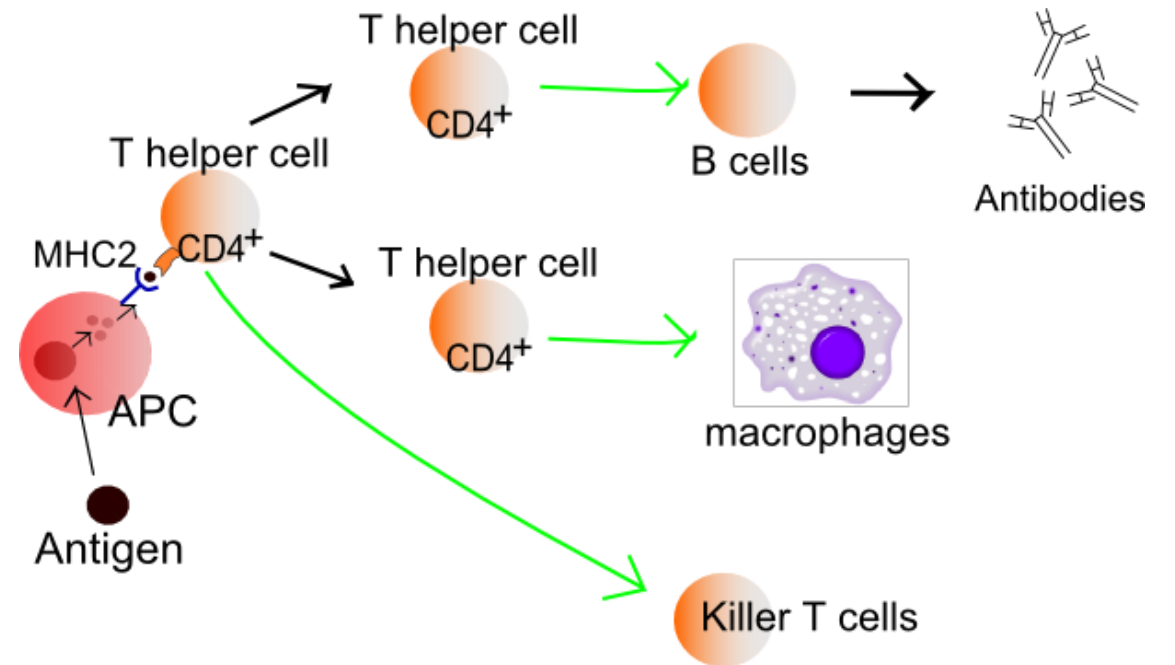
AUTOIMMUNE DISEASE

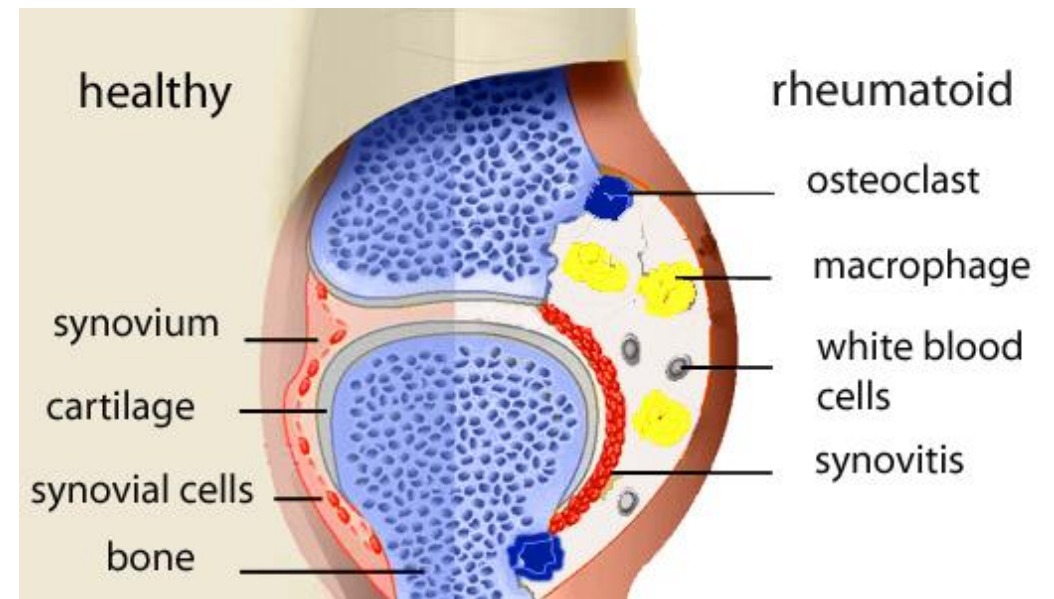
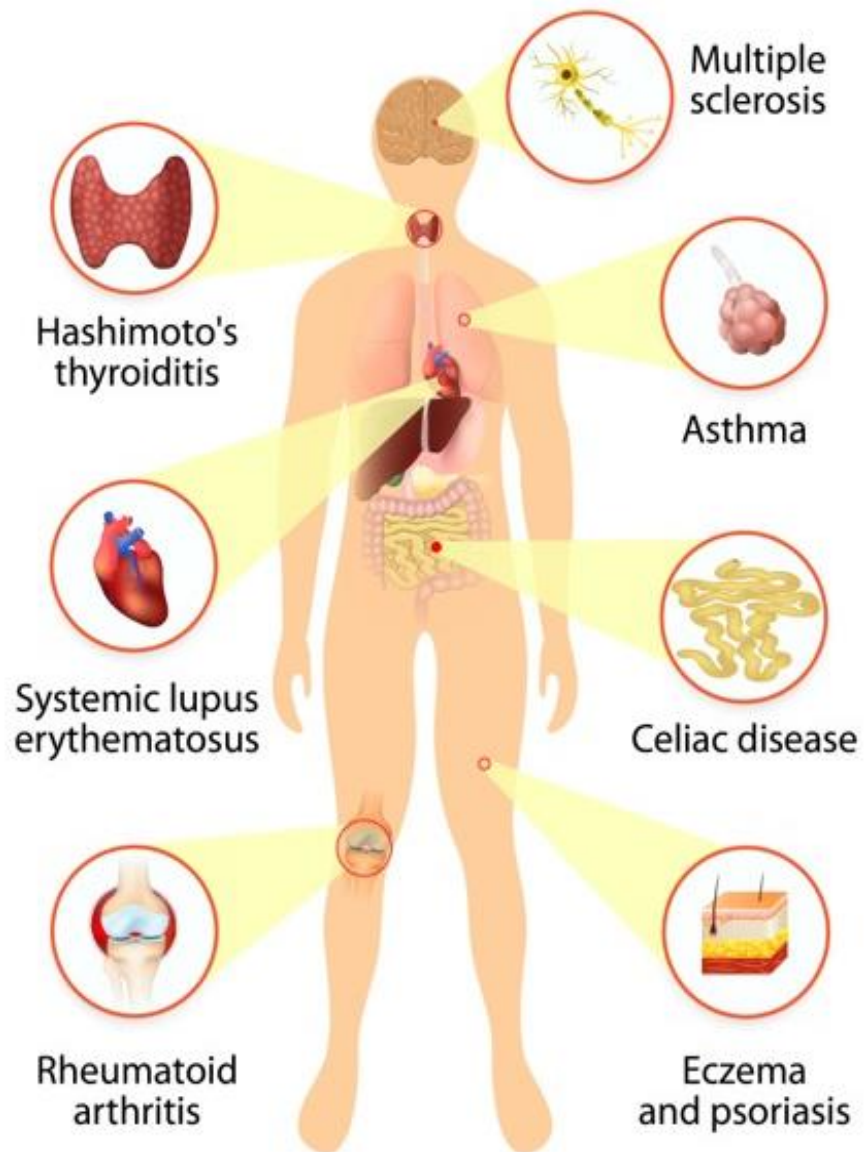
SUMMARY OF IMMUNE SYSTEM

Leucocytes



Processing of Antigen

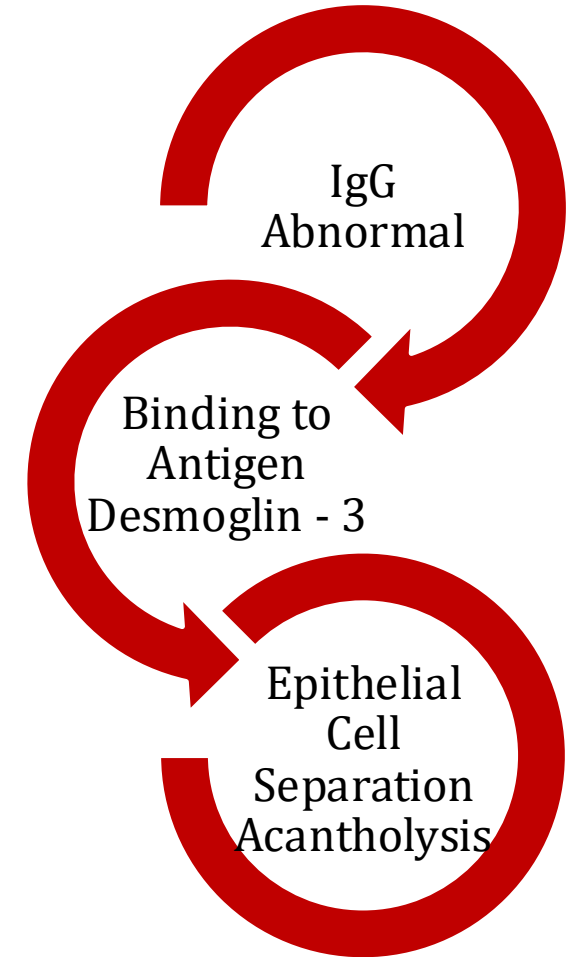
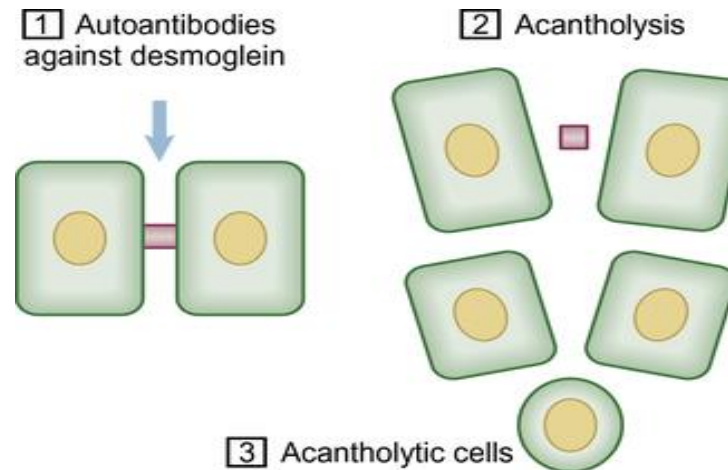
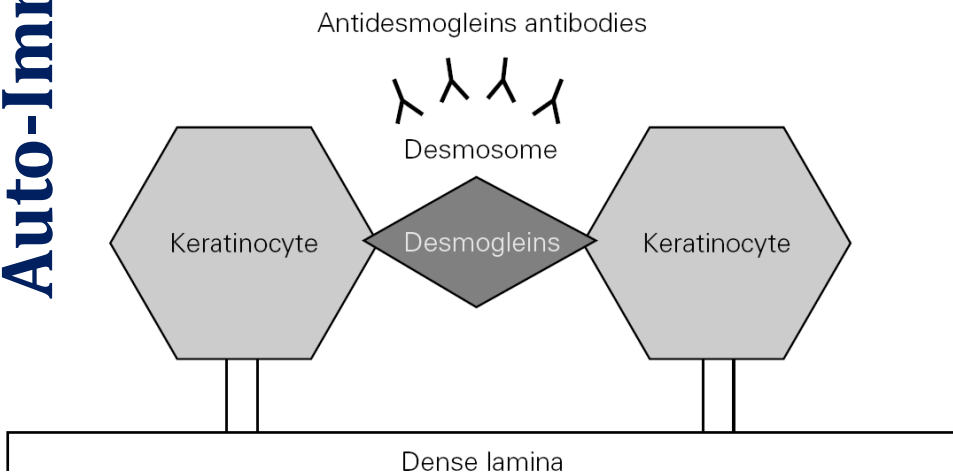
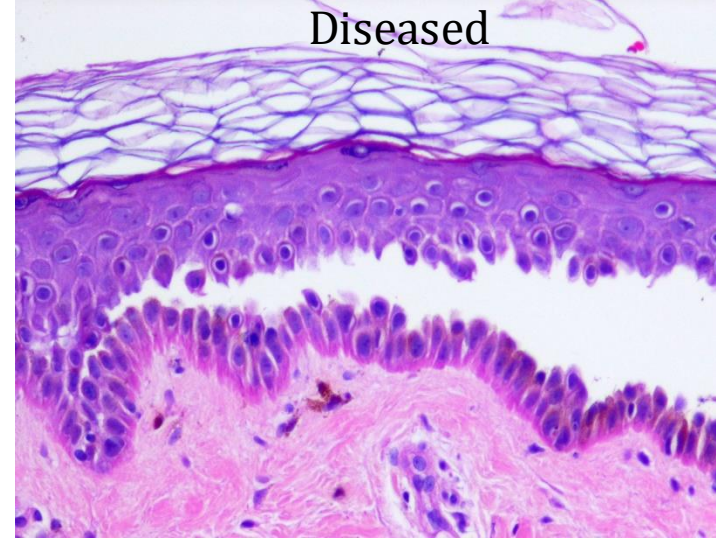
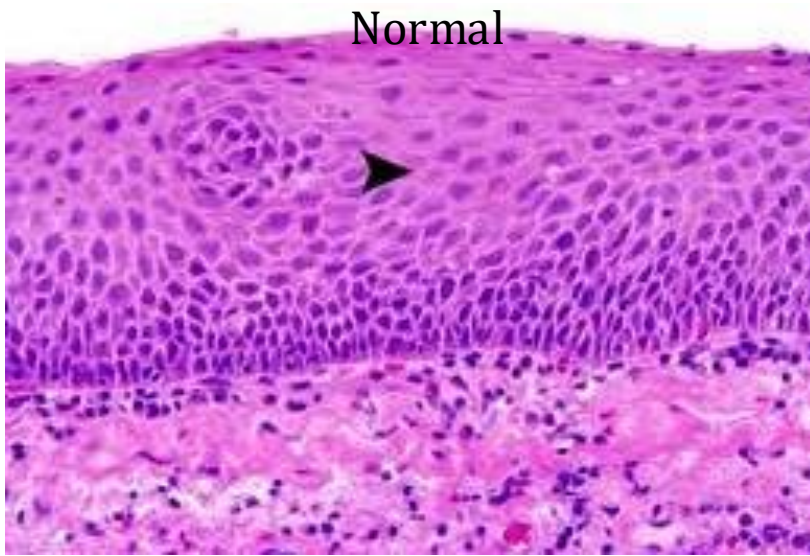




Advise: Healthy Lifestyle

- Nutritious Diet
- Physical Activity
- Healthy Sleep
- Stress-Free Life??

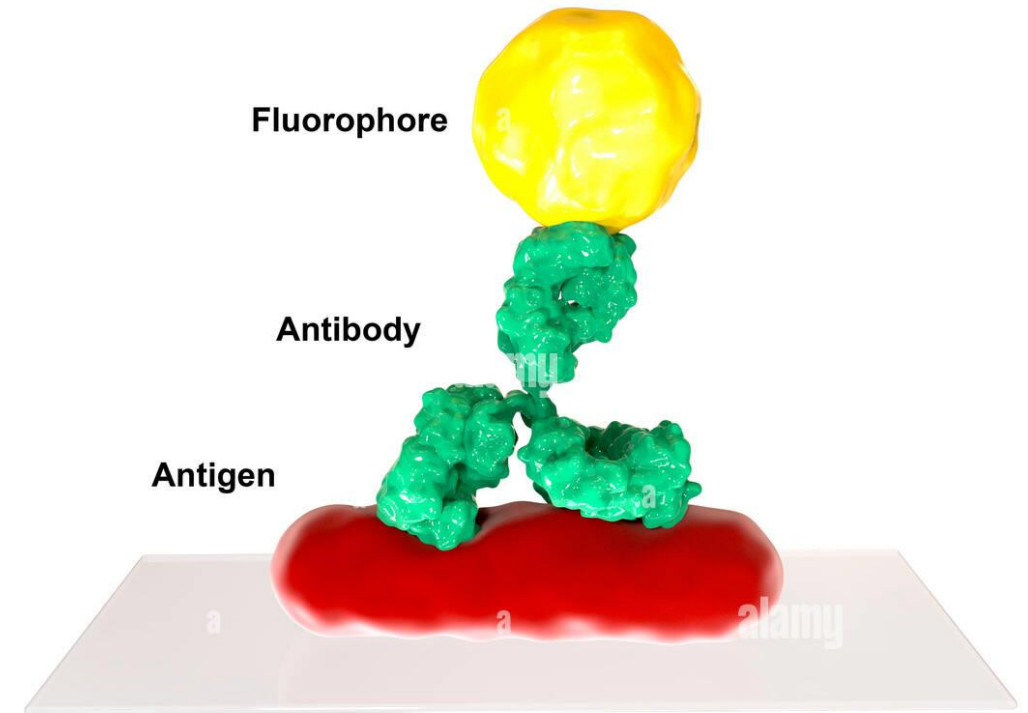
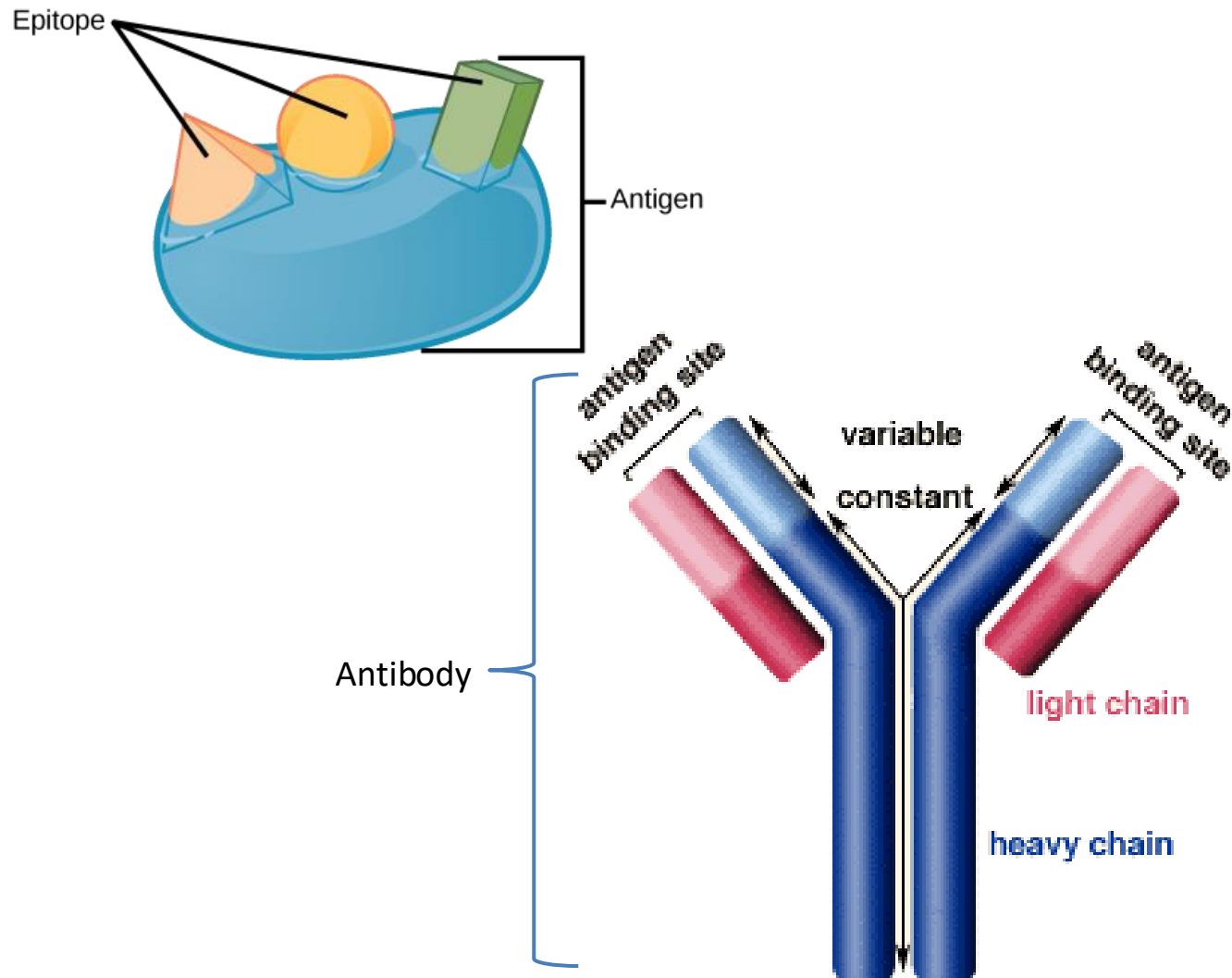
Example - Pemphigus



Immunofluorescence

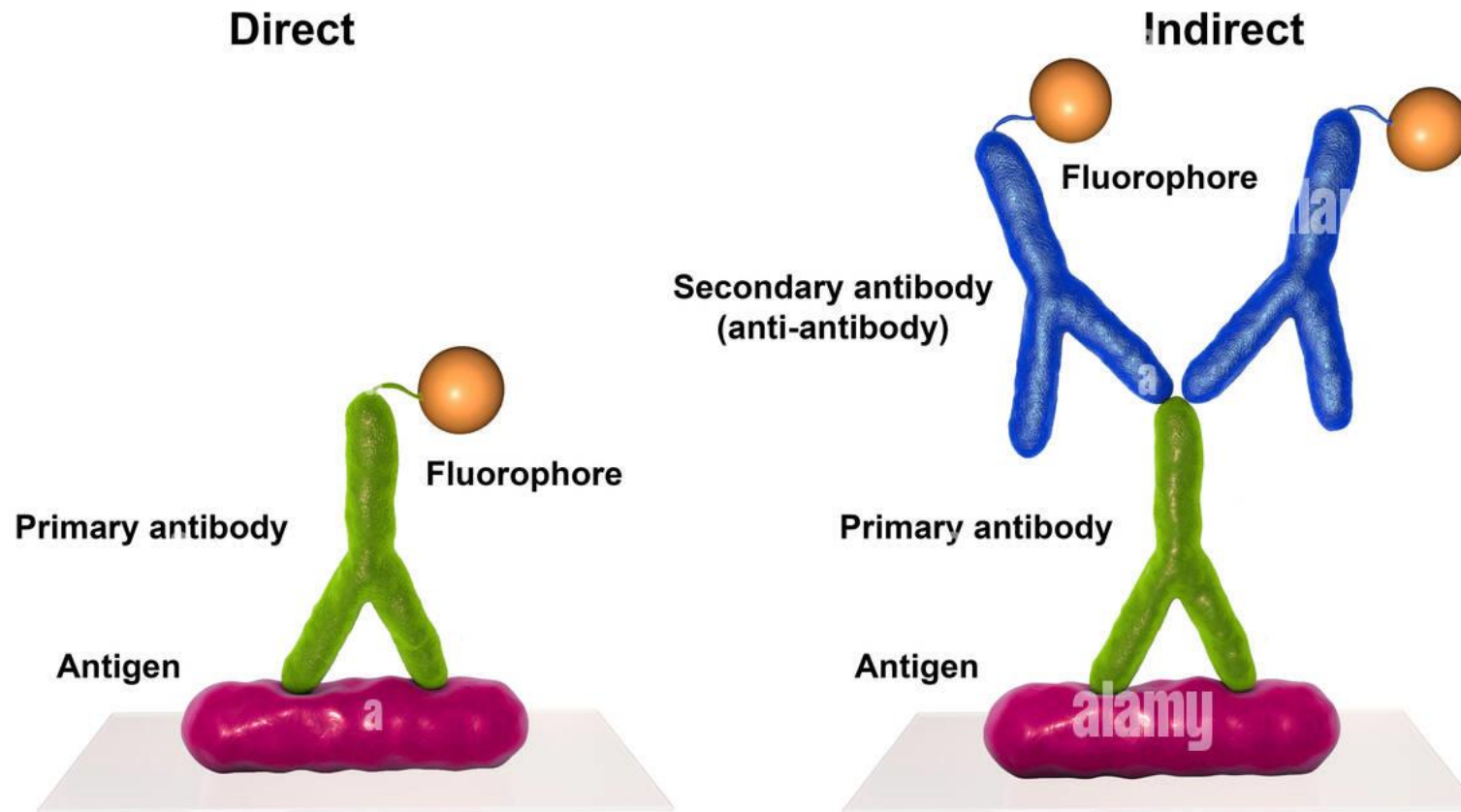
- If antibody molecules are tagged with a fluorescent dye, immune complexes containing these fluorescently labeled antibodies (FA) can be detected by colored light emission when excited by light of the appropriate wavelength.
- Antibody molecules bound to antigens in cells or tissue sections can similarly be visualized.
- The emitted light can be viewed with a fluorescence microscope, which is equipped with an UV light source.
- In this technique, known as immunofluorescence, fluorescent compounds such as:
 - fluorescein
 - rhodamine,
 - and phycoerythrin,
- are coupled to Abs which then can be detected by fluorescence microscope

Principles of Immunofluorescence

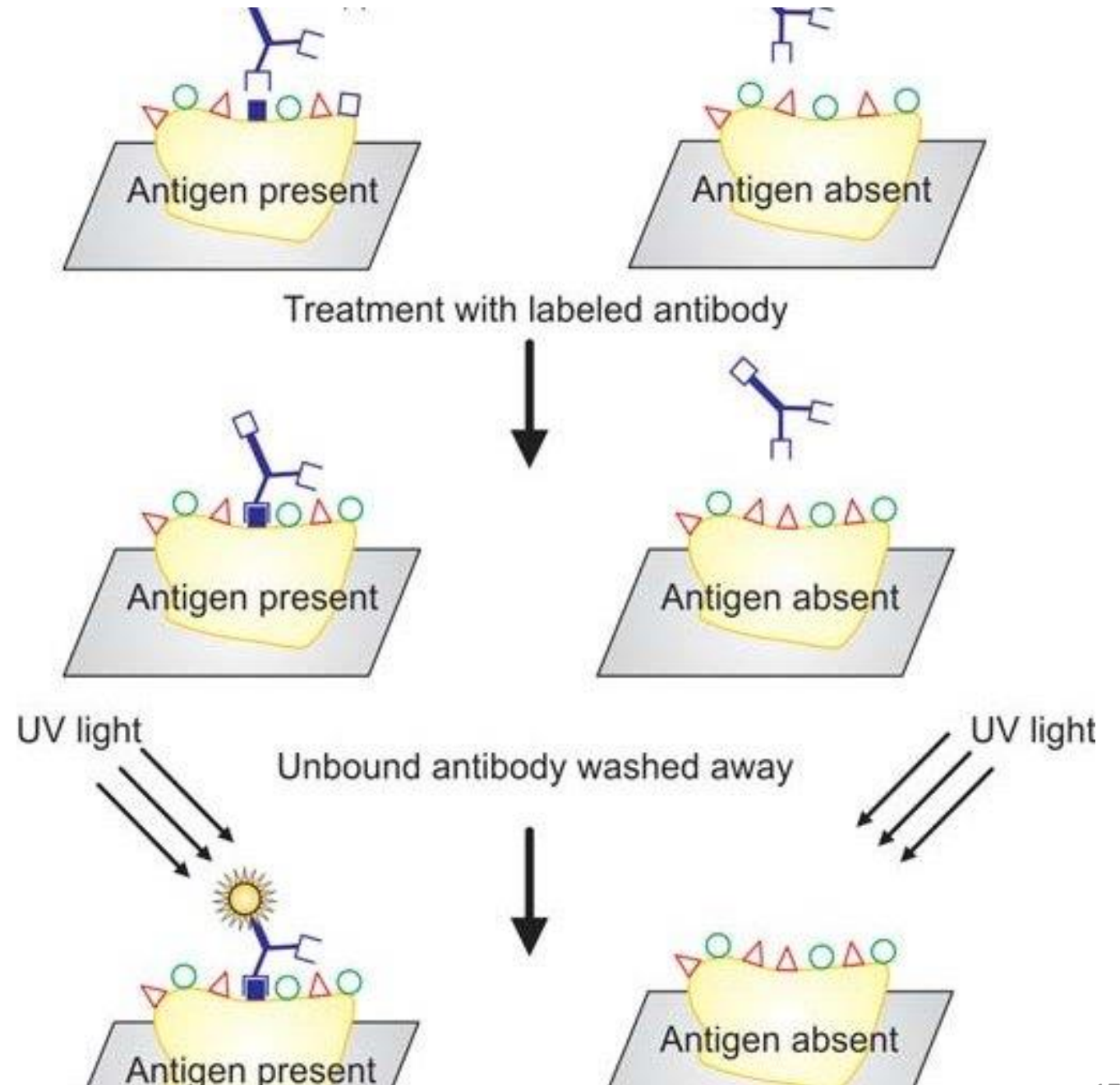


Principles of Immunofluorescence (Contd...)

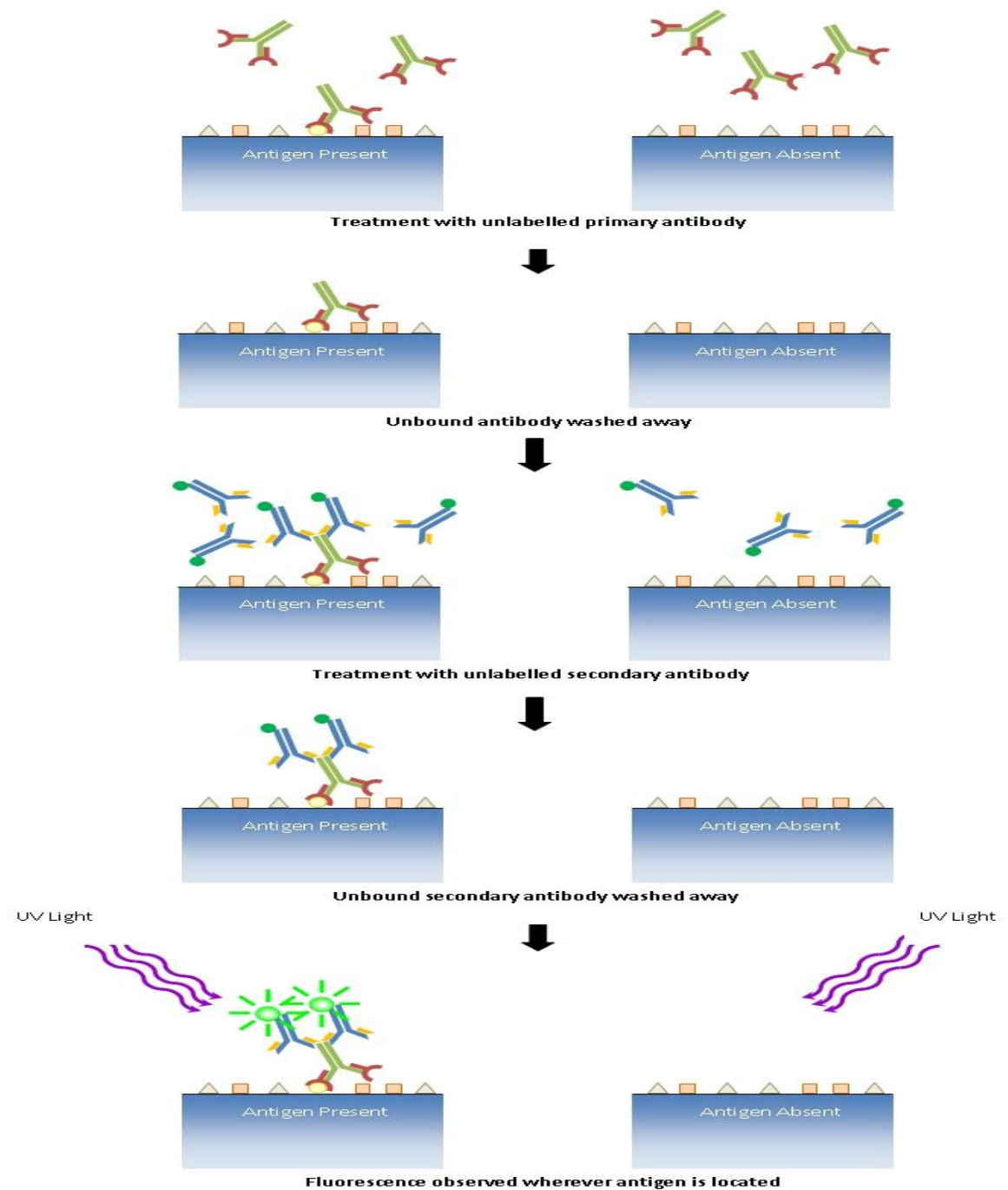
- IFA is a microscopic method that can detect and visualize the proteins expressed in cells via antigen—antibody reaction.



Procedure: Direct Immunofluorescence Process



Procedure: Indirect Immunofluorescence Process



Applications

Direct immunofluorescence

Skin/mucosa

Bullous pemphigoid

Pemphigus vulgaris

Lupus

Renal

Pulmonary transplantation

Indirect immunofluorescence

Pemphigus vulgaris

Paraneoplastic pemphigus

Bullous and/or cicatricial pemphigoid

Others fields of applications

Bone marrow

Cell surface markers

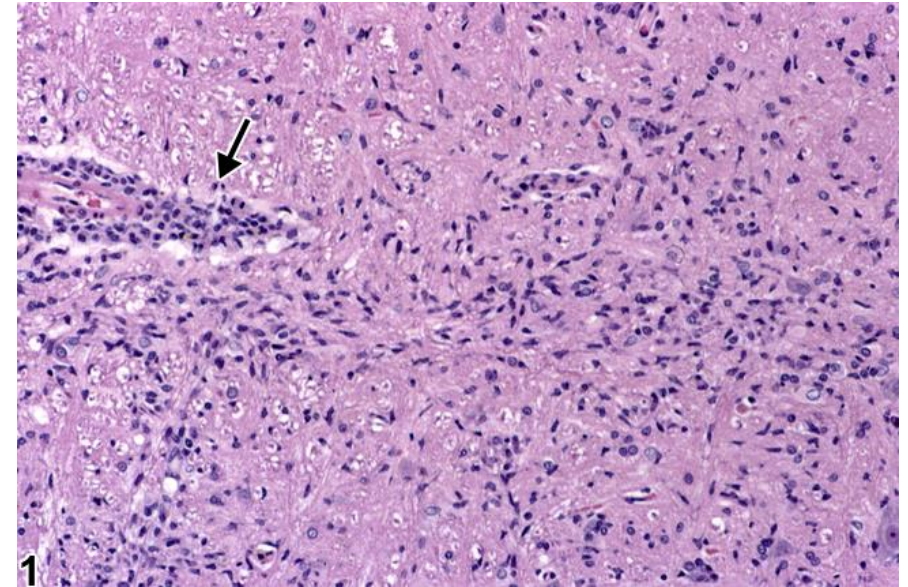
Infective disorders

Viral (Influenza virus, EBV, adeno virus, herpes virus, varicella zoster, CMV and HIV)

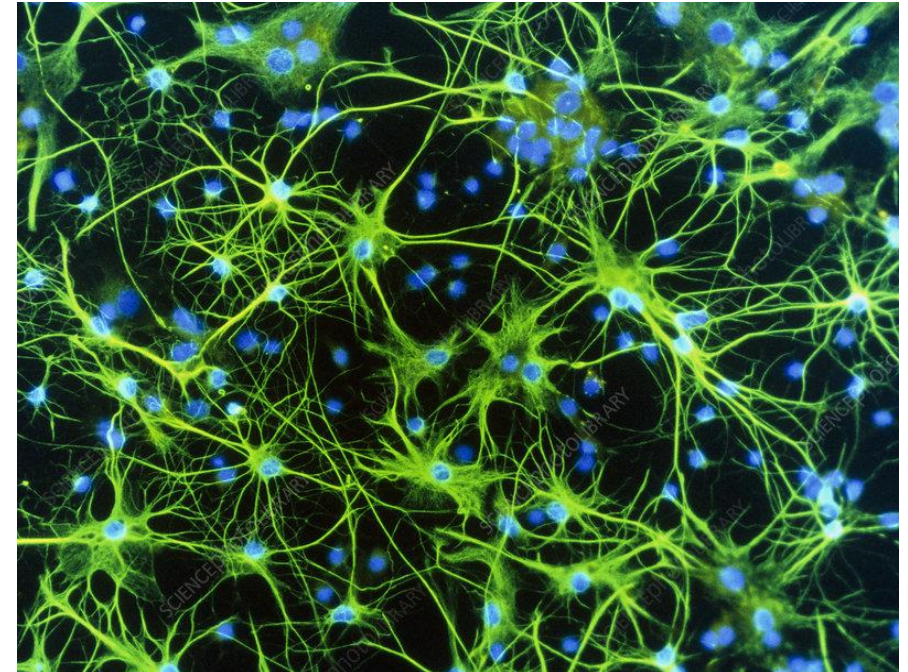
Bacterial (M. Pneumoniae, M. Tuberculosis, T. Pallidum)

Parasitic (Entamoeba, Giardia, Cyclospora)

Amyloid, Mucin, Lipid

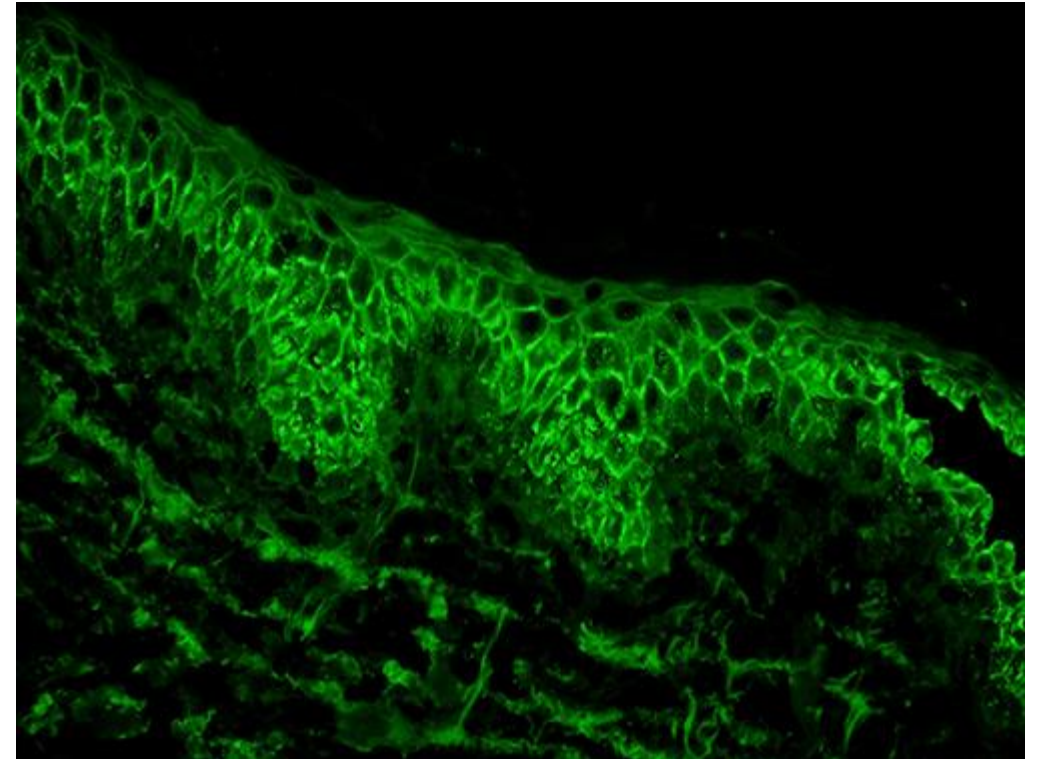
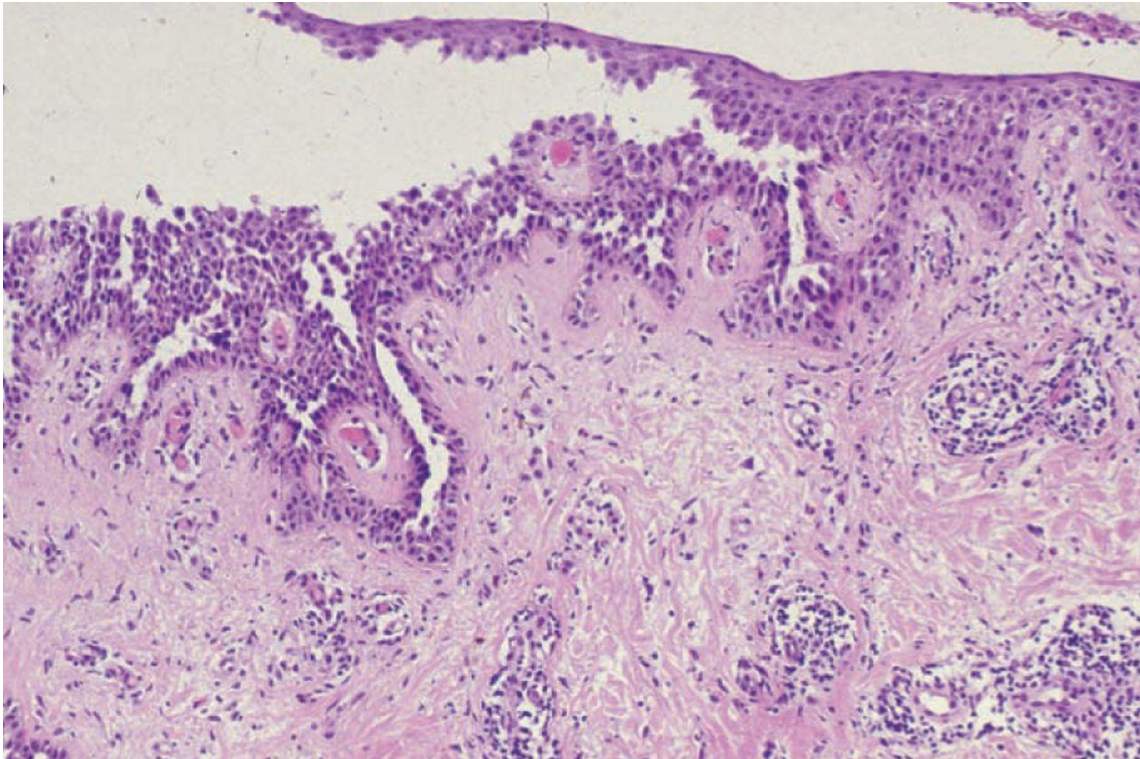


Asteroid cells of Brian



Applications

Vesiculobullous and Ulcerative Diseases Pemphigus Vulgaris



Limitations

- Quality and concentration of the antibody
- Proper handling of the specimen
- Choice of secondary antibodies
- Fluorophores undergo photobleaching as they are exposed to light, so they need to be stored in the dark and excited by the minimum light necessary for visualization

VESICULOBULLOUS DISEASES

CLASSIFICATION

Contd... Lecture – 1:

- **Viral Diseases**

1. Herpes Simplex Infection
2. Varicella Zoster Infection
3. Hand Foot and Mouth Disease
4. Herpangina
5. Rubeola (Measles)

- **Hereditary Disease**

1. Epidermolysis Bullosa

- **Immunologic Disease**

1. Pemphigus Vulgaris
2. Mucous Membrane Pemphigoid
3. Bullous Pemphigoid
4. Dermatitis Herpetiformis
5. Linear IgA - Bullous Disease

ULCERATIVE CONDITIONS

CLASSIFICATION

Contd... Lecture – 1:

- **Reactive Lesions**

1. Traumatic Ulceration

- **Bacterial Infections**

1. Tuberculosis
2. Syphilis
3. Actinomycosis

- **Fungal Infections**

1. Deep Fungal Infections
2. Subcutaneous Fungal Infections
3. Opportunistic Fungal Infections

- **Immunologic Disease**

1. Aphthous Ulcers
2. Reiter's Syndrome
3. Erythema Multiforme
4. Drug Reactions
5. Contact Allergies
6. Granulomatous Disease
7. Cyclic Neutropenia

- **Neoplasms**

1. Squamous Cell Carcinoma
2. Basal Cell Carcinoma

Reference

1. Oral Pathology – Clinico-Pathologic Correlations; Regezi, Sciubba & Jordan. 7th Edition; Elsevier – 2017.
 - Page 0-1 to 0-35
 - Page - 1
 - Page - 23
2. Immunofluorescence and its application in dermatopathology with oral manifestations: Revisited <https://www.jofs.in/article.asp?issn=0975-8844;year=2013;volume=5;issue=1;spage=2;epage=8;aulast=arvind>